Yanxi Chen

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Education

Tongji University *Shanghai, China*

Bachelor of Engineering (B.Eng.) in Intelligent Manufacturing Engineering *Sep. 2023 - Jun. 2027 (expected)*

- GPA: 3.88/4.0
- Major Courses: Control Engineering (A), Project of Mechatronics system (A), Sensors and Measurement Technology (A), Electronic Circuit Design (A), Fluid Mechanics and Hydraulic Transmission (A), Mechanics of Materials (A), Electrotechnology (Electrical Technology) (A), Matlab and Its Applications in Engineering (A), Python Programming (A), Database Technology and Applications (A), Artificial Intelligence (A).

Research Interests

My research interests lie in **medical robotics**, with a focus on **autonomous robotic systems that integrate learning-based control, mechanical design, and real-world experimental validation**. I aim to leverage biomechanics and computational methods to advance healthcare technologies, with particular emphasis on robustness, safety, and clinical applicability, and make meaningful contributions to human health.

Highlights: 3 years of programming experience; 2 years of mechanical design experience; 2 years of research experience in medical robotics with solid mathematical and theoretical background.

Publications

[A1] Learning Expert Strategy for Autonomous Robotic Endovascular Intervention via Decoupled Procedural Execution.
Yanxi Chen, Tianliang Yao, Shaolong Tang, Jiyuan Zhao, Hengyu Hu, Zhaoxing Li, Antonio J. Sánchez Egea, Peng Qi†.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Paper]

[A2] An On-demand Morphing Impeller Pump for Low-Speed, Hemocompatible pVADs.
Xiaolong Chen, Zhenyan Zhu, **Yanxi Chen**, Fulong Wang, Xu Chen, Liming Gao, Lingyun Shen, Yichao Tang†.
Advanced Healthcare Materials, 2026. [Paper]

[A3] Vision-Based Pose Perception for Interventional Robots via a Dual-Head U-Net Framework.
Yanxi Chen, Shunhan Ji, Zhongyu Yang, Shouwang Yang, Quan Zhang, Weihan Zhang, Zhangrui Yang, Jingqian Sun†, Yichao Tang†.
2026 IEEE Region 10 Symposium (TENSYMP), 2026. [Paper] [Video]

[A4] ALO–RATS: Accelerated, Long-Horizon Planning and Online Adjustment for Robot-Assisted Tissue Surgery.
Yanxi Chen*, Wuyue Xia*, Vincent Y. Wang, Patrick J. Codd, Leila J. Bridgeman, Ravi Prakash†.
IROS ELSR Workshop, 2026; extended manuscript in submission to *IEEE Robotics and Automation Letters (RA-L)*.

[A5] Maneuver-Aware Flow-Matching Action Chunking with Rolling-Horizon Replanning for Autonomous Robotic Endovascular Intervention.
Yanxi Chen, Tianliang Yao, Zhaoxing Li, Yuxiang Fan, Dong Liu, Tao Liu, Antonio J. Sánchez Egea, Peng Qi†.
IEEE Transactions on Automation Science and Engineering (T-ASE), under review, 2026.

[A6] Helically Wound Fiber Bragg Grating Sensor for Shape Sensing in Interventional Procedures.
Jing Lv, **Yanxi Chen**, Jingqian Sun, Chenlong Yang, Yichao Tang†, Lizhuang Sun†.
IEEE Sensors Journal, revised manuscript under review, 2026.

[A7] Passive moire-pattern imaging for large-field optical flatness assessment of textured bipolar plates.
Liyan Zhao, **Yanxi Chen**, Yuxuan Lu, Weimin Zhang†, Shaoyu Lin, Wanying Liu, Hailang Wan, Christopher Ehrmann, Junying Min.
Optics and Lasers in Engineering, revised manuscript under review, 2026.

[A8] LAMP: Learning Action-Relevant Manipulation-Centric Visual Perception with Illumination-Adaptive Latent Representations.
Wentong Tian, Tianliang Yao, Jiyuan Zhao, Zhengyu Shi, **Yanxi Chen**, Peng Qi†.
IEEE International Conference on Robotics and Automation (ICRA), under review, 2027.

*Equal contribution. †Corresponding author.

Research Experiences

Georgia Institute of Technology *Remote and Atlanta, USA*

Research Assistant at BioMedical Mechatronics Lab, advised by Prof. Yue Chen; mentored by Yilin Cai and Zhefeng Huang *Jul 2026 – Present*

- **Research Topics:** autonomous surgical robotics, vision-language-action models, shared autonomy, and sim-to-real transfer.
- Developed a three-arm dVRK-Si simulation framework in Isaac Lab for collaborative robotic surgery.
- Fine-tuned and adapted SmolVLA for autonomous third-arm manipulation, and developed a language-guided shared-autonomy framework combining VLA task grounding, learned policy execution, and safety-constrained low-level control for surgical assistance.

Duke University

Remote

Research Assistant at Brain Tool Lab, advised by Prof. Patrick Codd and Prof. Leila Bridgeman; mentored by Dr. Ravi Prakash

May 2026 – Sep. 2026

- **Research Topics:** autonomous robotic laser surgery, safe motion planning, control theory.
- Developed ALO–RATS, a 3D volumetric laser-resection planner that constructs complete treatment sequences and jointly refines laser position, orientation, and energy using deterministic multiscale optimization and terminal whole-plan evaluation **[A4]**.

Tongji University

Shanghai, China

Research Assistant at Medical Robotics Lab, advised by Prof. Peng Qi; mentored by Tianliang Yao

Sep. 2025 – Sep. 2026

- **Research Topics:** autonomous endovascular robotics, guidewire navigation, reinforcement learning, embodied intelligence.
- Designed a hierarchical learning framework for autonomous endovascular guidewire navigation, integrating high-level policy learning with predictive optimal control constraints for safety-critical execution; validated task-level autonomy in complex vascular simulation environments; conducted sim-to-real validation on a physical endovascular robotic platform **[A1]**.
- Developed a flow-matching based policy for guidewire navigation, learning reusable local action sequences for challenging vascular maneuvers and integrating them with global feedback control **[A5]**.

Tongji University

Shanghai, China

Research Assistant at BioNACH Lab, advised by Prof. Yichao Tang

Dec. 2023 – Jul. 2025

- **Research Topics:** medical robotics, interventional robotic systems, medical device design.
- Designed a novel impeller blade structure and fabricated it using reverse molding and 3D printing techniques **[A2]**.
- Developed and optimized an interventional robotic system for cardio-cerebrovascular procedures, integrating mechanical design, rapid prototyping, vision-based pose perception algorithms, control system implementation, and experimental validation **[A3]**.

Additional Experiences

KTUR Summer School 2025

Switzerland, Germany, France

Project Team Lead [LINK]

Aug. 2025

- Selected as the **first Chinese participant** in the KTUR Summer School program; led a student team (*Nanovoyage*) to explore commercialization pathways for a medical robotics research project, including market analysis, technology positioning, and investor-oriented pitch presentations.

Fuel Cell Bipolar Plate Inspection and Handling Systems

Shanghai, China

Team member

May 2025 – Sep. 2025

- Developed an image-based moiré-pattern inspection framework for large-field bipolar-plate flatness assessment, integrating experimental image acquisition with ZEISS contour measurements as ground truth for method validation **[A7]**.
- Designed a new vacuum-assisted handling and assembly mechanism for large-format fuel cell components, including CAD drawing, 3D mechanical modeling, and refinement of the device structure for fabrication and patent documentation.
- Contributed to **two Chinese patent applications** covering moiré-based bipolar-plate flatness sensing and vacuum-assisted component handling.

Research on Innovative Design of a Lightweight Collaborative Wheeled-Legged Robot

Shanghai, China

Robot Designer

Mar. 2025 – May 2025

- Replicated a wheeled-legged mobile robot by integrating a lightweight robotic manipulator with a two-wheeled locomotion system, aiming to enhance mobility and operational flexibility in complex environments through manipulation–locomotion coupling.

Honors & Awards (Selected)

2026 **IEEE/RSJ IROS 2026 Travel Support Grant**, Awarded $1,500 in competitive travel support.

2024&2025 **Tongji University Excellent Student Scholarship**, top students in Tongji University.

2024&2025 **Social Activity Scholarship**, awarded for sustained extracurricular involvement.

2025 **Mechanical Engineering Innovation and Creativity Competition**, National Second Prize.

2024 **Tuoya Scholarship**, awarded to students with outstanding proficiency in English.

2024 **Tongji University Applied Mechanics Innovation Competition**, Third Prize.

2024 **VEX Robotics Competition at Tongji University**, Second Prize and Design Award.

2023 **National Innovation Engineering Competition**, National Third Prize.

Skills

Programming Python, LaTeX, SQL, C#, MATLAB, ROS.

Frameworks PyTorch, Isaac Lab/Sim, SOFA, OpenCV, Git, Anaconda.

Mechanical Tools SolidWorks, AutoCAD, 3ds Max, 3D print.

Languages Chinese, English.